Exercise Sheet 3 – Solution

Even though this PVK is structured into thematic sections and some content has not yet been covered, it is explicitly permitted and *should* be the case that in the problem sets *all the material* covered in the semester and in the lecture may be used.

Exercise 1
Differentiate the following functions.

$$(1)\ e^x\left(1-\sin^2(x)\right) \qquad (2)\ \sin\left(e^{x^2}\right) \qquad (3)\ \frac{e^x+1}{e^x-1} \qquad (4)\ x\cos(3^x)$$

Solution:

(1)
$$e^x\left(1-2\cos(x)\sin(x)-\sin^2(x)\right)$$

(2)
$$2xe^{x^2}\cos\left(e^{x^2}\right)$$

(3)
$$\frac{-2e^x}{(e^x-1)^2}$$

(4)
$$\cos(3^x)-3^x x\log(3)\sin(3^x)$$

Exercise 2 (FS2023 G. Felder)
Prove: Let f be a convex function $f:\mathbb{R}\to\mathbb{R}$, such that $f(x)\leq 2023$ for all $x\in\mathbb{R}$. Then f is constant.

Solution: For a convex function f on $\mathbb{R}$ we have

$$\frac{f(c)-f(b)}{c-b}\geq\frac{f(b)-f(a)}{b-a} \quad \text{for all } a<b<c.$$

Let f be convex and not constant. Then there are two points $x<y\in\mathbb{R}$ such that $f(x)\neq f(y)$, i.e., $f(x)<f(y)$ or $f(x)>f(y)$. In the first case set $a=x$, $b=y$. Then $\frac{f(b)-f(a)}{b-a}>0$ and

$$f(c)\geq f(b)+(c-b)\frac{f(b)-f(a)}{b-a}\to\infty \quad (c\to\infty),$$

so $f(c)>2023$ for sufficiently large c. In the second case, i.e., if $f(x)>f(y)$, set $b=x$ and $c=y$ and write the inequality as

$$\frac{f(b)-f(c)}{c-b}\leq\frac{f(a)-f(b)}{b-a} \quad \text{for all } a<b<c.$$

Then $\frac{f(b)-f(c)}{c-b}\geq 0$ and

$$f(a)\geq f(b)+(b-a)\frac{f(b)-f(c)}{c-b}\to\infty \quad (a\to-\infty),$$

contradicting the assumption that $f(a) \leq 2023$ for all a.

Exercise 3
Let I be an interval and $f : I \to \mathbb{R}$ a differentiable function. Show that f is convex if and only if f' is monotonically increasing.

Solution: $\implies$
First assume that f' is monotonically increasing and show that f must be convex. Let $a < b$ and $x \in (a, b)$. By the Mean Value Theorem, we know there exist $\xi \in (a, x)$ and $\zeta \in (x, b)$ such that

$$f'(\xi) = \frac{f(x) - f(a)}{x - a}, \qquad f'(\zeta) = \frac{f(b) - f(x)}{b - x}$$

Because f' is monotonically increasing, we have: $\xi \leq \zeta \implies \frac{f(x)-f(a)}{x-a} \leq \frac{f(b)-f(x)}{b-x}$. Hence f is convex.

$\impliedby$

Now assume that f is convex. Let $a < b \in I$ and $h > 0$ such that $a + h < b - h$. h should thus be smaller than the mean of a and b, so that h fits twice between a and b. We apply a definition of convexity to the interval $(a, b - h)$ with $x = a + h$:

$$\frac{f(a + h) - f(a)}{h} \leq \frac{f(b - h) - f(a + h)}{(b - h) - (a + h)}$$

and to the interval $(a + h, b)$ with $x = b - h$.

$$\frac{f(b - h) - f(a + h)}{(b - h) - (a + h)} \leq \frac{f(b) - f(b - h)}{h}$$

Comparing the two inequalities we obtain:

$$\frac{f(a + h) - f(a)}{h} \leq \frac{f(b) - f(b - h)}{h}$$

Considering the limit as $h \to 0^+$, both terms become derivatives. For the right term it is not directly obvious. It follows via a simple transformation with $\tilde{h} = -h$:

$$\frac{f(b) - f(b - h)}{h} = \frac{f(b) - f(b + \tilde{h})}{-\tilde{h}} = \frac{f(b + \tilde{h}) - f(b)}{\tilde{h}}.$$

Thus in the limit $h \to 0^+$ we obtain: $f'(a) \leq f'(b)$. Since $a < b$ were chosen arbitrarily, the theorem follows.

Exercise 4
Show that every continuous function $f : [a, b] \to \mathbb{R}$ is Riemann-integrable.

Solution: Since the interval $[a, b]$ is compact, we know that f is uniformly continuous. So for every $\varepsilon > 0$ there exists a uniform $\delta > 0$ such that:

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$$

for all $x, y \in [a, b]$.

Now we choose a partition of $[a, b]$ as

$$a = x_0 < x_1 < \ldots < x_{n-1} < x_n = b \qquad\qquad x_k - x_{k-1} < \delta.$$

We now define for $k \in \{1, \ldots, n\}$:

$$c_k = \min\{f(x) \mid x_{k-1} \leq x \leq x_k\} \qquad\qquad d_k = \max\{f(x) \mid x_{k-1} \leq x \leq x_k\}.$$

The maxima and minima are well-defined because each interval is compact and f is continuous.

Now define the lower and upper step functions for $x \in [a, b]$ as:

$$\ell(x) = \begin{cases} c_k & \text{if } x \in [x_{k-1}, x_k) \\ c_n & \text{if } x = b \end{cases} \qquad\qquad u(x) = \begin{cases} d_k & \text{if } x \in [x_{k-1}, x_k) \\ d_n & \text{if } x = b \end{cases}$$

We have $\ell \leq f \leq u$ and due to the uniform continuity of f, we have $d_k - c_k < \varepsilon$, because we chose $x_k - x_{k-1} < \delta$. Therefore:

$$\int_a^b (u - \ell)(x)dx = \sum_{k=1}^{n} \underbrace{(d_k - c_k)}_{<\varepsilon}(x_k - x_{k-1}) < \varepsilon \sum_{k=1}^{n} (x_k - x_{k-1}) = \varepsilon(b - a)$$

and the theorem follows, since $\varepsilon > 0$ was chosen arbitrarily.

Exercise 5
Let (f_n) with $f_n : [a, b] \to \mathbb{R}$ be a sequence of integrable functions that converges uniformly to a function $f : [a, b] \to \mathbb{R}$. Prove, that f is integrable and

$$\int_a^b f(x)\, dx = \lim_{n\to\infty} \int_a^b f_n(x)\, dx.$$

Solution: Fix $\varepsilon > 0$. By uniform convergence, there exists N such that $|f_n - f| \leq \varepsilon$ on $[a, b]$ for all $n \geq N$.
Since f_N is integrable, there exist step functions ℓ, u with $\ell \leq f_N \leq u$ and $\int_a^b (u - \ell)\, dx < \varepsilon$. Set $\hat{\ell} = \ell - \varepsilon$ and $\hat{u} = u + \varepsilon$. Then both $\hat{\ell}$ and $\hat{u}$ are step functions. Also, since $|f_N - f| < \varepsilon$,

$$\hat{\ell} = \ell - \varepsilon \leq f_N - \varepsilon \leq f \leq f_N + \varepsilon \leq u + \varepsilon = \hat{u}$$

and (because $\hat{u} - \hat{\ell} = u - \ell + 2\varepsilon$)

$$\int_a^b (\hat{u} - \hat{\ell})\, dx = \int_a^b (u - \ell)\, dx + 2\varepsilon(b - a) < \varepsilon + 2\varepsilon(b - a).$$

As $\varepsilon > 0$ is arbitrary, we find that f is integrable.
Moreover, using monotonicity of the Riemann Integral and the triangle inequality we find,

$$\left| \int_a^b f\, dx - \int_a^b f_n\, dx \right| = \left| \int_a^b (f - f_n)\, dx \right| \leq \int_a^b |f - f_n|\, dx \leq \varepsilon(b - a) \quad \forall n \geq N.$$

Exercise 6 (FS2023 G. FELDER)
Compute the limit of Riemann sums

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \sin\left(\frac{k}{n}\right).$$

Solution: We recognize that this is the definition of the Riemann integral of $\sin(x)$ on a uniform partition of $[0,1]$. (*Tip: For this, insert the limits $k=1$ and $k=n$.* This shows the interval.) Therefore this limit can be computed by the following integral

$$\int_0^1 \sin(x)dx = -\cos(1)+\cos(0) = 1-\cos(1).$$

Exercise 7
Let $f:[a,b]\to\mathbb{R}$ be a continuous function. Show that for every constant $C\in\mathbb{R}$ the function $F:[a,b]\to\mathbb{R}$ is an antiderivative of f and every antiderivative has the following form with a constant $C\in\mathbb{R}$:

$$F(x)=\int_a^x f(t)dt + C.$$

Solution: Since f is continuous, we know that f is integrable on $[a,b]$. We first want to show that $F(x)$ is an antiderivative of f. We know, due to the continuity of f, that

$$|x-x_0|<\delta \implies |f(x)-f(x_0)|<\varepsilon$$

Now consider the δ-interval $(x_0, x_0+\delta)\cap[a,b]$.

$$\begin{aligned}
\left|\frac{F(x)-F(x_0)}{x-x_0}-f(x_0)\right| &= \left|\frac{1}{x-x_0}\left(\int_a^x f(t)dt-\int_a^{x_0} f(t)dt\right)-f(x_0)\right| \\
&= \left|\frac{1}{x-x_0}\left(\int_{x_0}^x f(t)dt\right)-\frac{1}{x-x_0}\int_{x_0}^x f(x)dt\right| \\
&= \left|\frac{1}{x-x_0}\left(\int_{x_0}^x f(t)-f(x_0)dt\right)\right| \\
&\le \frac{1}{x-x_0}\int_{x_0}^x \underbrace{|f(t)-f(x_0)|}_{<\varepsilon}dt \\
&< \varepsilon\,\frac{1}{x-x_0}\int_{x_0}^x dt = \varepsilon
\end{aligned}$$

Next, the δ-interval $(x_0-\delta, x_0)\cap[a,b]$.

$$\begin{aligned}
\left|\frac{F(x)-F(x_0)}{x-x_0}-f(x_0)\right| &= \left|\frac{1}{x-x_0}\left(\int_a^x f(t)dt-\int_a^{x_0} f(t)dt\right)-f(x_0)\right| \\
&= \left|\frac{1}{x-x_0}\left(-\int_x^{x_0} f(t)dt\right)-\frac{1}{x-x_0}\int_{x_0}^x f(x_0)dt\right| \\
&= \left|\frac{1}{x_0-x}\left(\int_x^{x_0} f(t)-f(x_0)dt\right)\right| \\
&\le \frac{1}{x_0-x}\int_x^{x_0} \underbrace{|f(t)-f(x_0)|}_{<\varepsilon}dt \\
&< \varepsilon\,\frac{1}{x-x_0}\int_x^{x_0} dt = \varepsilon
\end{aligned}$$

Since $\varepsilon>0$ was chosen arbitrarily, we have: $\lim_{x\to x_0}\frac{F(x)-F(x_0)}{x-x_0}=F'(x_0)=f(x_0)$. Thus $F(x_0)$ is

an antiderivative of f. The uniqueness up to the constant C follows:

$$\left(F(x) - \int_a^x f(t)dt\right)' = f(x) - f(x) = 0,$$

which means the only difference lies in a constant. The theorem follows.

Exercise 8 (FS2024 A. FIGALLI)
Let $f : [0,1] \to \mathbb{R}$ be a continuously differentiable function.

(a) Suppose there exists an $x_0 \in [0,1]$ such that $f(x_0) = 0$. Show that

$$|f(x)| \leq \int_0^1 |f'(s)|ds \qquad \forall x \in [0,1].$$

(b) Suppose that $\int_0^1 f(s)ds = 0$. Show that

$$|f(x)| \leq \int_0^1 |f'(s)|ds \qquad \forall x \in [0,1]$$

and conclude that

$$\int_0^1 |f(s)|ds \leq \int_0^1 |f'(s)|ds.$$

(c) Let $m = \int_0^1 f(s)ds \in \mathbb{R}$. Show that

$$\int_0^1 |f(s) - m|ds \leq \int_0^1 |f'(s)|ds.$$

Solution:

(a) We can apply the Fundamental Theorem and obtain

$$|f(x)| = |f(x) - f(x_0)| = \left|\int_x^{x_0} f'(s)ds\right| \leq \int_x^{x_0} |f'(s)|ds \leq \int_0^1 |f'(s)|ds.$$

Here the last estimate follows from the fact that $|f'(s)|$ is positive. Thus the Riemann integral must be monotonically increasing when we extend the integration domain.

(b) For the integral to be zero, f cannot be only positive or only negative. Therefore, by the Intermediate Value Theorem, there must exist an $x_0 \in [0,1]$ such that $f(x_0) = 0$. Hence we can apply (a) and obtain the first inequality.
From this we conclude by integrating both sides with respect to s

$$\int_0^1 |f(s)|ds \leq \int_0^1 \int_0^1 |f'(s)|ds\, ds = \int_0^1 |f'(s)|ds.$$

(c) We apply (b) to the function $g(s) = f(s) - m$, because we know that $m = \int_0^1 f(s)ds$. For we know that

$$\int_0^1 g(s)ds = \int_0^1 f(s)ds - \int_0^1 mds = \int_0^1 f(s)ds - \int_0^1 \int_0^1 f(s)ds\, ds = \int_0^1 f(s)ds - \int_0^1 f(s)ds = 0.$$

The inequality from (b) yields, knowing that m is constant:

$$\int_0^1 |f(s) - m|ds \leq \int_0^1 |f'(s) - m'|ds = \int_0^1 |f'(s)|ds.$$